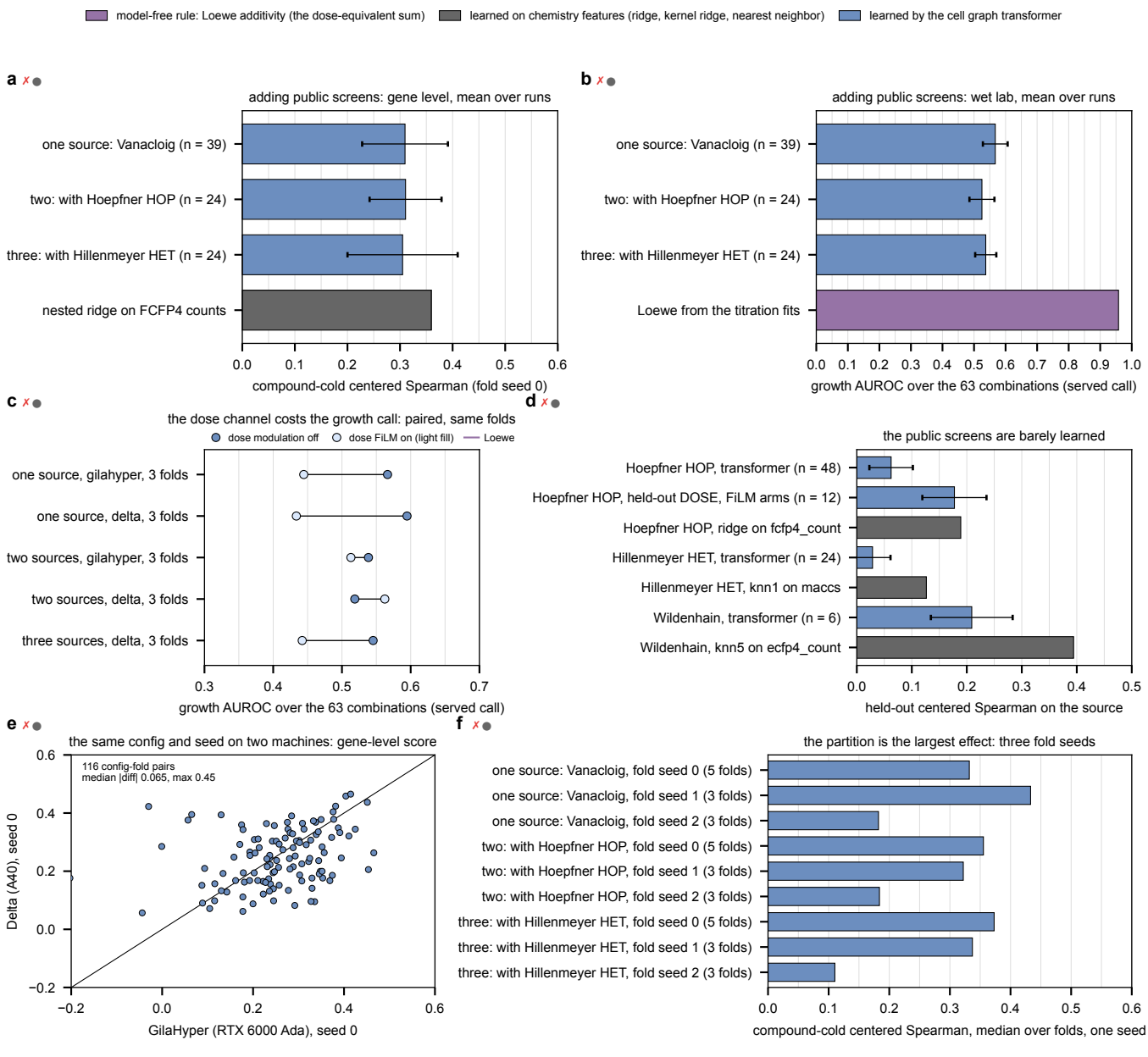

model-free rule: Loewe additivity (the dose-equivalent sum)
learned on chemistry features (ridge, kernel ridge, nearest neighbor)
learned by the cell graph transformer
a
adding public screens: gene level, mean over runs
one source: Vanacloig (n = 39)
two: with Hoepfner HOP (n = 24)
three: with Hillenmeyer HET (n = 24)
nested ridge on FCFP4 counts
0.0 0.1 0.2 0.3 0.4 0.5 0.6
compound-cold centered Spearman (fold seed 0)
b
adding public screens: wet lab, mean over runs
one source: Vanacloig (n = 39)
two: with Hoepfner HOP (n = 24)
three: with Hillenmeyer HET (n = 24)
Loewe from the titration fits
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1.0
growth AUROC over the 63 combinations (served call)
c
the dose channel costs the growth call: paired, same folds
dose modulation off
dose FiLM on (light fill)
Loewe
one source, gilahyper, 3 folds
one source, delta, 3 folds
two sources, gilahyper, 3 folds
two sources, delta, 3 folds
three sources, delta, 3 folds
0.3 0.4 0.5 0.6 0.7
growth AUROC over the 63 combinations (served call)
d
the public screens are barely learned
Hoepfner HOP, transformer (n = 48)
Hoepfner HOP, held-out DOSE, FiLM arms (n = 12)
Hoepfner HOP, ridge on fcfp4_count
Hillenmeyer HET, transformer (n = 24)
Hillenmeyer HET, knn1 on maccs
Wildenhain, transformer (n = 6)
Wildenhain, knn5 on ecfp4_count
0.0 0.1 0.2 0.3 0.4 0.5
held-out centered Spearman on the source
e
the same config and seed on two machines: gene-level score
116 config-fold pairs
median |diff| 0.065, max 0.45
Delta (A40), seed 0
GilaHyper (RTX 6000 Ada), seed 0
0.6 0.4 0.2 0.0 −0.2
−0.2 0.0 0.2 0.4 0.6
f
the partition is the largest effect: three fold seeds
one source: Vanacloig, fold seed 0 (5 folds)
one source: Vanacloig, fold seed 1 (3 folds)
one source: Vanacloig, fold seed 2 (3 folds)
two: with Hoepfner HOP, fold seed 0 (5 folds)
two: with Hoepfner HOP, fold seed 1 (3 folds)
two: with Hoepfner HOP, fold seed 2 (3 folds)
three: with Hillenmeyer HET, fold seed 0 (5 folds)
three: with Hillenmeyer HET, fold seed 1 (3 folds)
three: with Hillenmeyer HET, fold seed 2 (3 folds)
0.0 0.1 0.2 0.3 0.4 0.5 0.6
compound-cold centered Spearman, median over folds, one seed